

Sequence analysis tools for viruses and structural RNAs

Eric Nawrocki

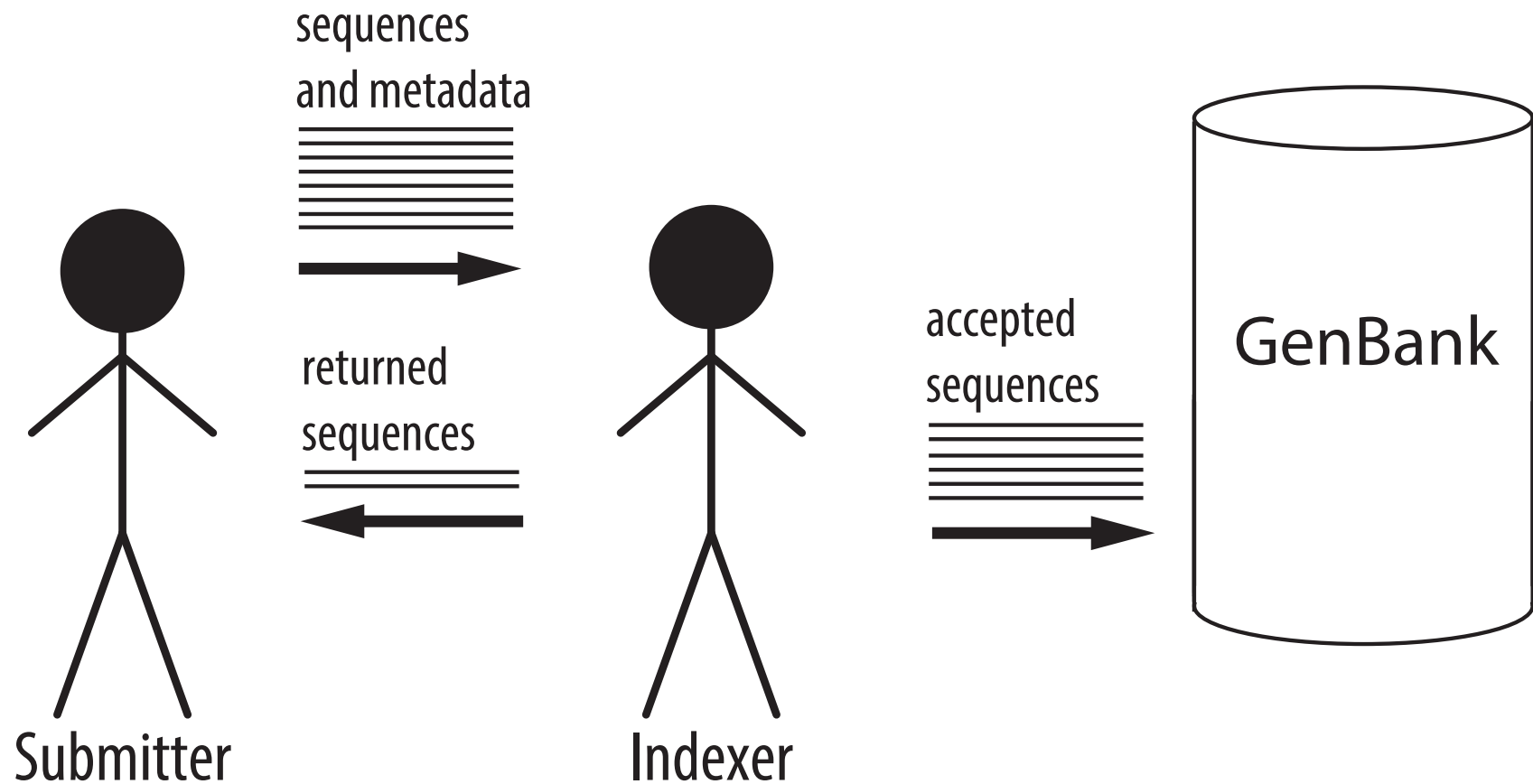
Intramural Research Program
National Library of Medicine
National Institutes of Health



Two main areas of my research:

1. Viral sequence analysis tools, since 2015
2. Structural RNA analysis tools, since 2004

GenBank indexers handle incoming sequence submissions



SOFTWARE

Open Access

VADR: validation and annotation of virus sequence submissions to GenBank

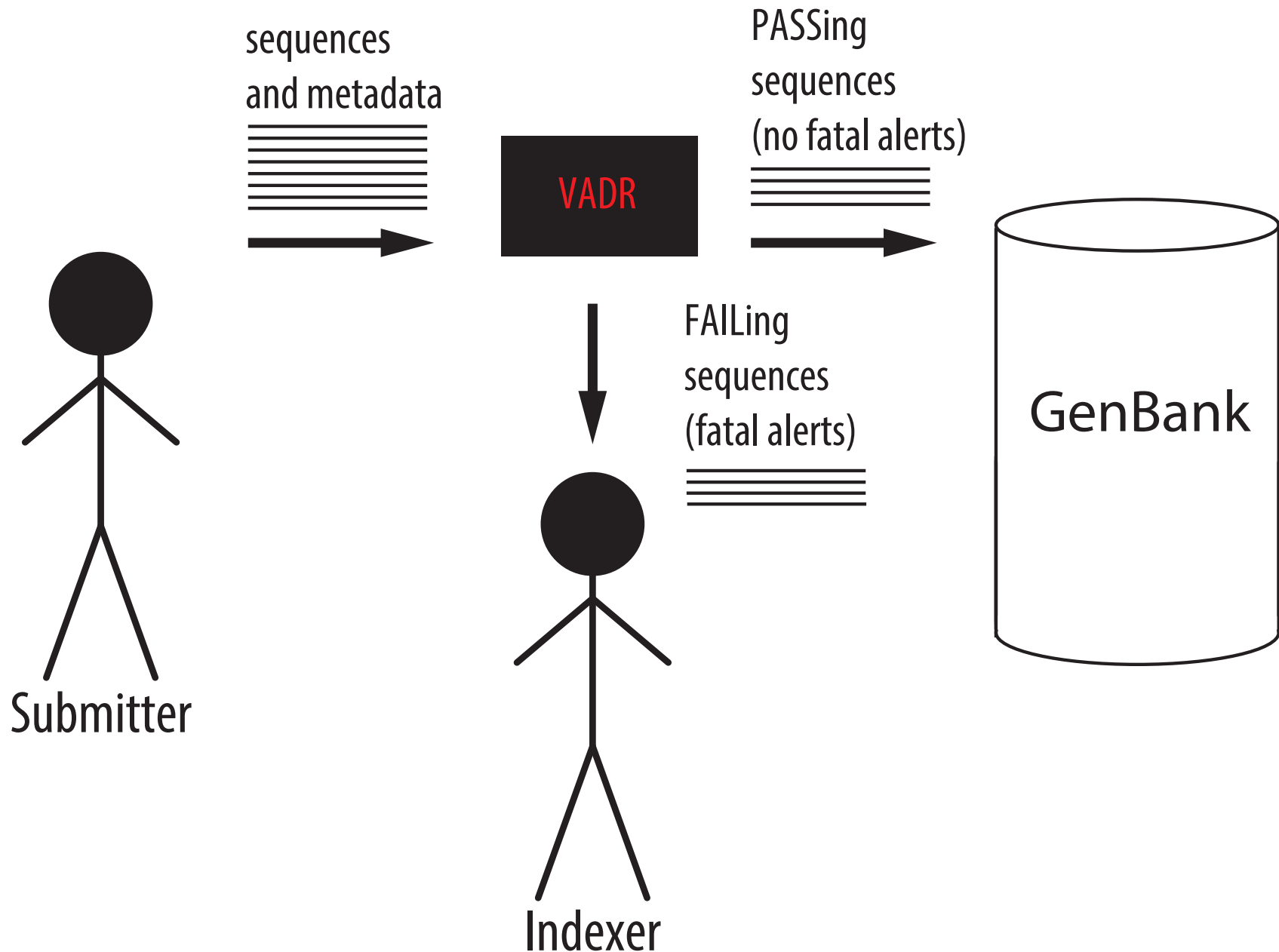


Alejandro A. Schäffer^{1,2}, Eneida L. Hatcher², Linda Yankie², Lara Shonkwiler^{2,3}, J. Rodney Brister², Ilene Karsch-Mizrachi² and Eric P. Nawrocki^{2*} 

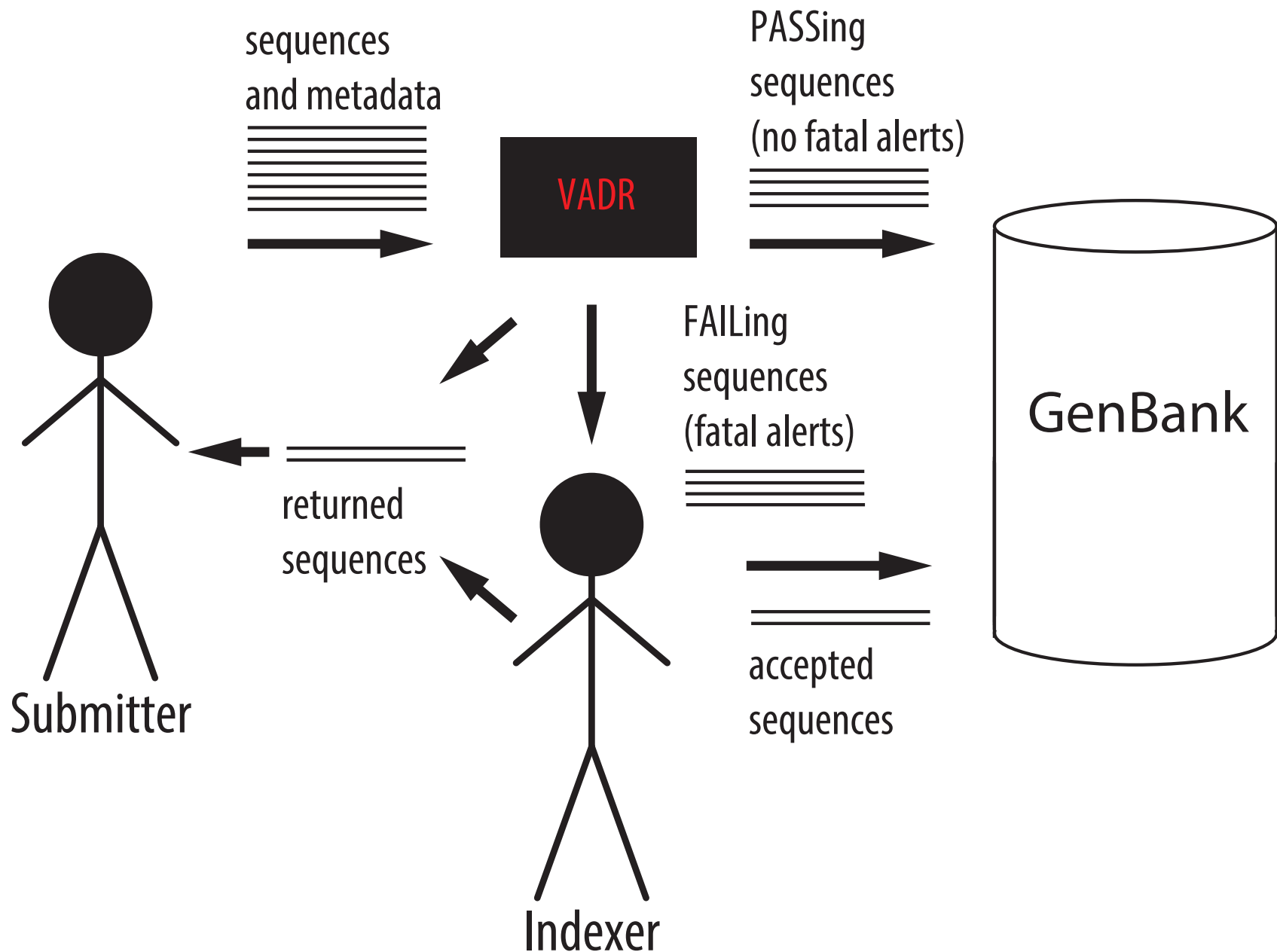
- general tool for reference-based annotation of viral sequences
- used for Norovirus and Dengue virus submissions since 2018
- used for SARS-CoV-2 submissions since March 2020
- also used manually for submissions of RSV, Mpox, measles, Influenza, and more

VADR assists GenBank indexers:

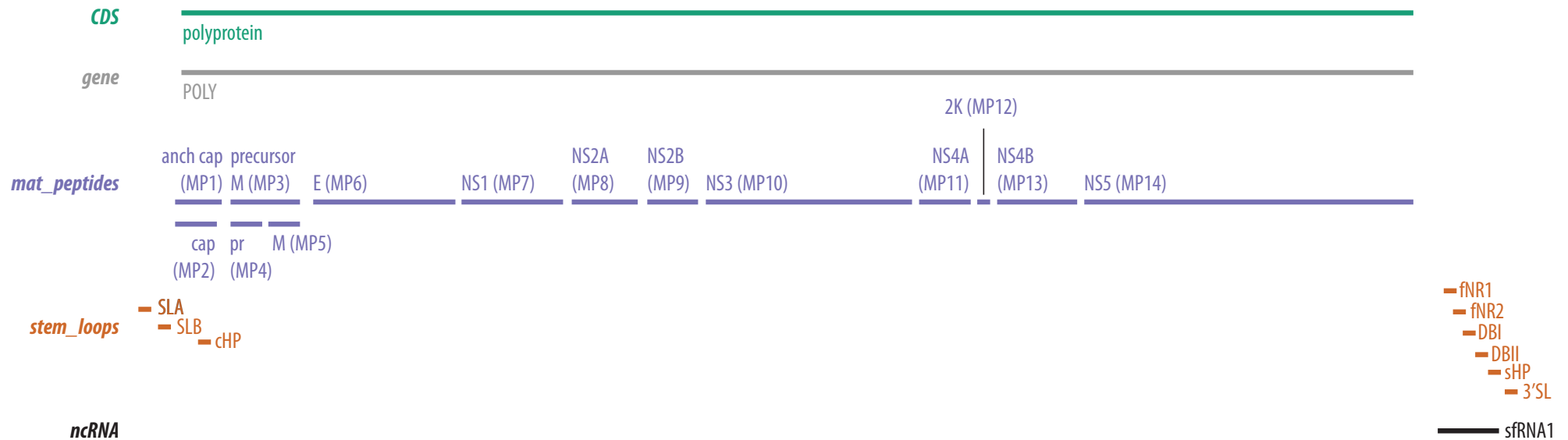
Each sequence **PASSes** or **FAILs**



Indexers decide fate of some **FAILing** sequences
but some are sent directly back to submitter with error reports



VADR builds a reference model of a RefSeq and its features



NC_001477 MODEL



Group: Dengue; Subgroup: 1

VADR validates and annotates each input sequence using its best-matching model

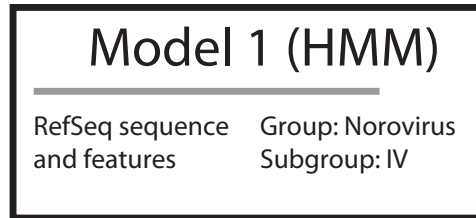
- Each sequence S proceeds through 4 stages:
 1. **Classification**
 2. **Coverage determination**
 3. **Alignment**
 4. **Protein validation**

Different types of alerts are identified and reported at each stage

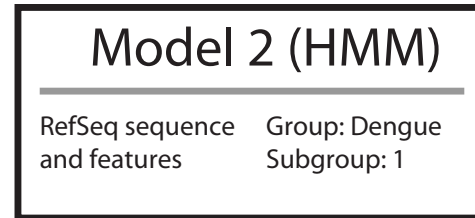
Stage 1: Classification

Score each sequence
with all models
(HMMER3 shortened pipeline)

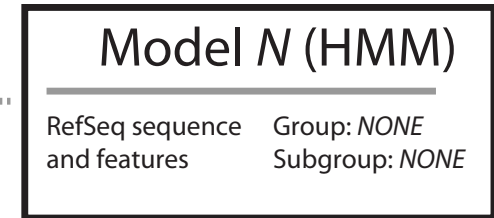
input sequences:



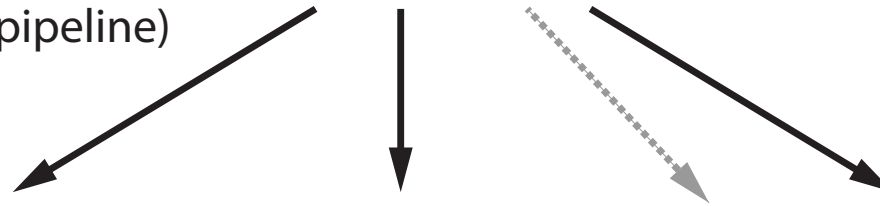
low HMM score



highest HMM score



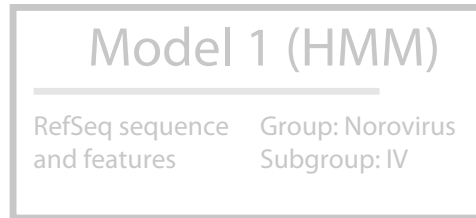
low HMM score



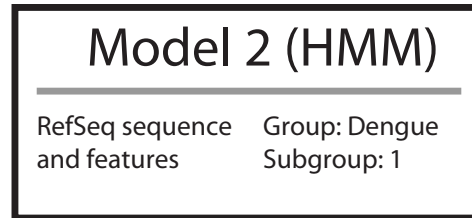
Stage 1: Classification

Score each sequence
with all models
(HMMER3 shortened pipeline)

input sequences:

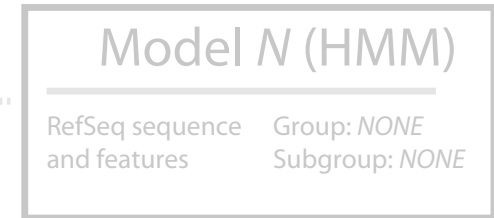


low HMM score



highest HMM score

**best-matching model
used in remaining stages**

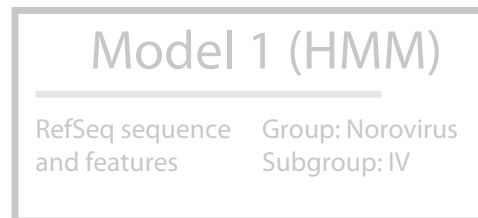


low HMM score

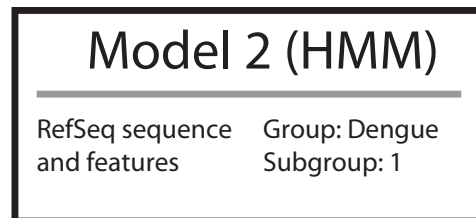
Stage 1: Classification

Score each sequence
with all models
(HMMER3 shortened pipeline)

input sequences:

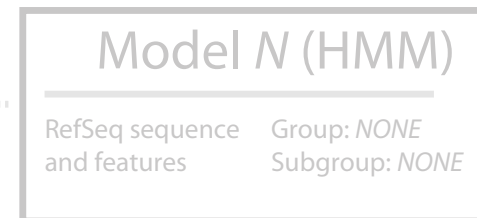


low HMM score



highest HMM score

**best-matching model
used in remaining stages**



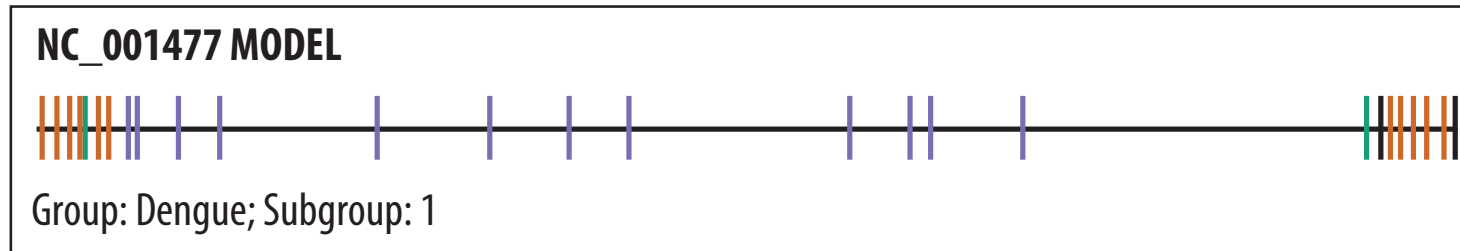
low HMM score

code	S/F	error message	description
Fatal alerts detected in the classification stage			
noannotn*	S	NO_ANNOTATION	no significant similarity detected
revcompl*	S	REVCOMPLEM	sequence appears to be reverse complemented
incsbgrp	S	INCORRECT_SPECIFIED_SUBGROUP	score difference too large between best overall model and best specified subgroup model
incgroup	S	INCORRECT_SPECIFIED_GROUP	score difference too large between best overall model and best specified group model
Non-fatal alerts detected in the classification stage			
qstsbgrp	S	QUESTIONABLE_SPECIFIED_SUBGROUP	best overall model is not from specified subgroup
qstgroup	S	QUESTIONABLE_SPECIFIED_GROUP	best overall model is not from specified group
indfclas	S	INDEFINITE_CLASSIFICATION	low score difference between best overall model and second best model (not in best model's subgroup)
lowscore	S	LOW_SCORE	score to homology model below low threshold

Stage 2: Coverage determination

Search each sequence with best-matching model (HMMER3 full pipeline)

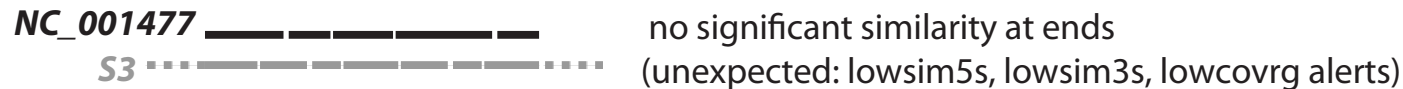
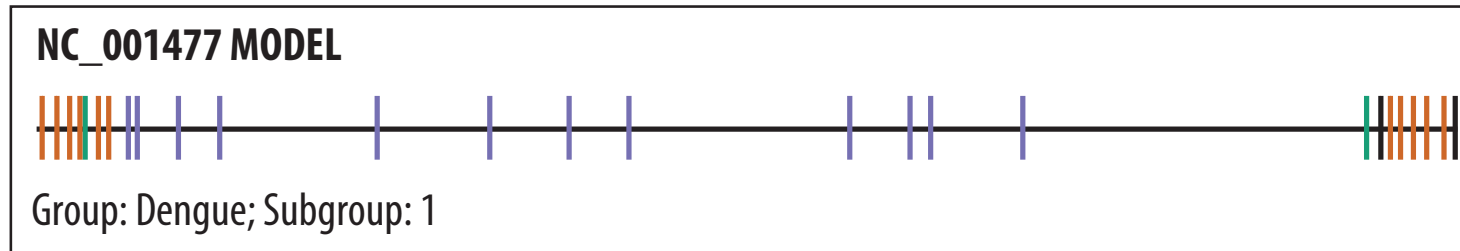
input sequences that match best to NC_001477:



Stage 2: Coverage determination

Search each sequence with best-matching model (HMMER3 full pipeline)

input sequences that match best to NC_001477:

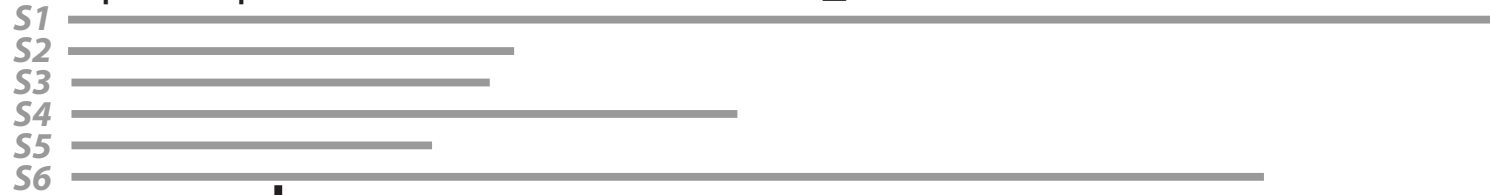


code	S/F	error message	description
Fatal alerts detected in the coverage stage			
lowcovrg	S	LOW_COVERAGE	low sequence fraction with significant similarity to homology model
dupregin	S	DUPLICATE_REGIONS	similarity to a model region occurs more than once
discontn	S	DISCONTINUOUS_SIMILARITY	not all hits are in the same order in the sequence and the homology model
indfstrn	S	INDEFINITE_STRAND	significant similarity detected on both strands
lowsim5s	S	LOW_SIMILARITY_START	significant similarity not detected at 5' end of the sequence
lowsim3s	S	LOW_SIMILARITY_END	significant similarity not detected at 3' end of the sequence
lowsimis	S	LOW_SIMILARITY	internal region without significant similarity
Non-fatal alerts detected in the coverage stage			
biasdseq	S	BIASED_SEQUENCE	high fraction of score attributed to biased sequence composition

Stage 3: Alignment and feature mapping

Align each sequence to its best-matching model (Infernal's cmalign)

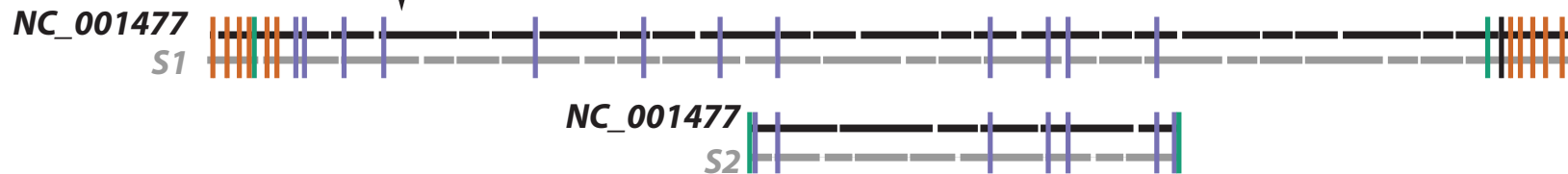
input sequences that match best to NC_001477:



NC_001477 MODEL

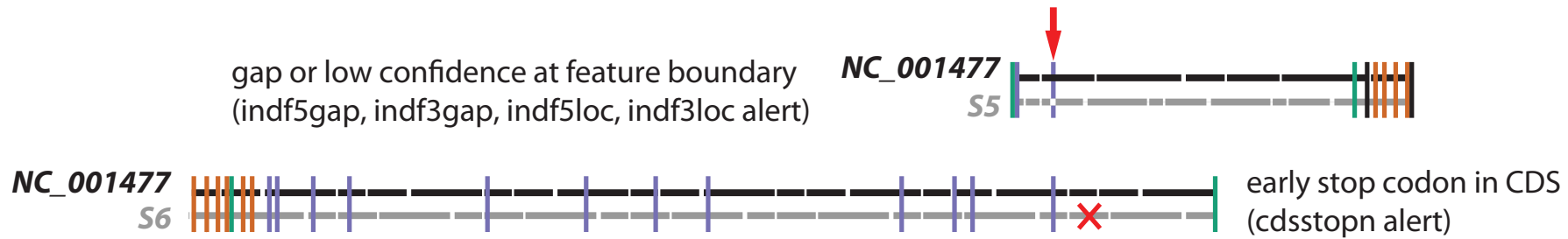


Group: Dengue; Subgroup: 1



Stage 3: Alignment and feature mapping

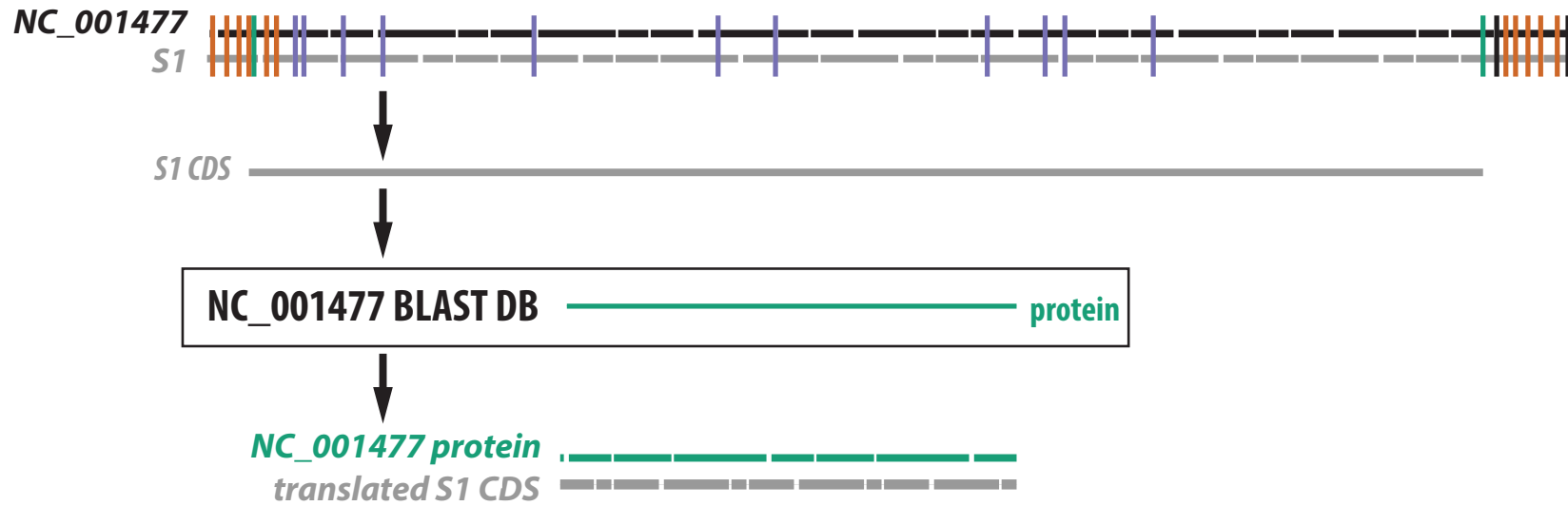
Align each sequence to its best-matching model (Infernal's cmalign)



code	S/F	error message	description
Fatal alerts detected in the annotation stage			
unexdivg*	S	UNEXPECTED_DIVERGENCE	sequence is too divergent to confidently assign nucleotide-based annotation
noftrann*	S	NO_FEATURES_ANNOTATED	sequence similarity to homology model does not overlap with any features
mutstart	F	MUTATION_AT_START	expected start codon could not be identified
mutendcd	F	MUTATION_AT_END	expected stop codon could not be identified, predicted CDS stop by homology is invalid
mutendns	F	MUTATION_AT_END	expected stop codon could not be identified, no in-frame stop codon exists 3' of predicted valid start codon
mutendex	F	MUTATION_AT_END	expected stop codon could not be identified, first in-frame stop codon exists 3' of predicted stop position
unexleng	F	UNEXPECTED_LENGTH	length of complete coding (CDS or mat_peptide) feature is not a multiple of 3
cdsstopn	F	CDS_HAS_STOP_CODON	in-frame stop codon exists 5' of stop position predicted by homology to reference
peptrans	F	PEPTIDE_TRANSLATION_PROBLEM	mat_peptide may not be translated because its parent CDS has a problem
pepadjcy	F	PEPTIDE_ADJACENCY_PROBLEM	predictions of two mat_peptides expected to be adjacent are not adjacent
indfantn	F	INDEFINITE_ANNOTATION	nucleotide-based search identifies CDS not identified in protein-based search
indf5gap	F	INDEFINITE_ANNOTATION_START	alignment to homology model is a gap at 5' boundary
indf5loc	F	INDEFINITE_ANNOTATION_START	alignment to homology model has low confidence at 5' boundary
indf3gap	F	INDEFINITE_ANNOTATION_END	alignment to homology model is a gap at 3' boundary
indf3loc	F	INDEFINITE_ANNOTATION_END	alignment to homology model has low confidence at 3' boundary
lowsim5f	F	LOW_FEATURE_SIMILARITY_START	region within annotated feature at 5' end of sequence lacks significant similarity
lowsim3f	F	LOW_FEATURE_SIMILARITY_END	region within annotated feature at 3' end of sequence lacks significant similarity
lowsimif	F	LOW_FEATURE_SIMILARITY	region within annotated feature lacks significant similarity

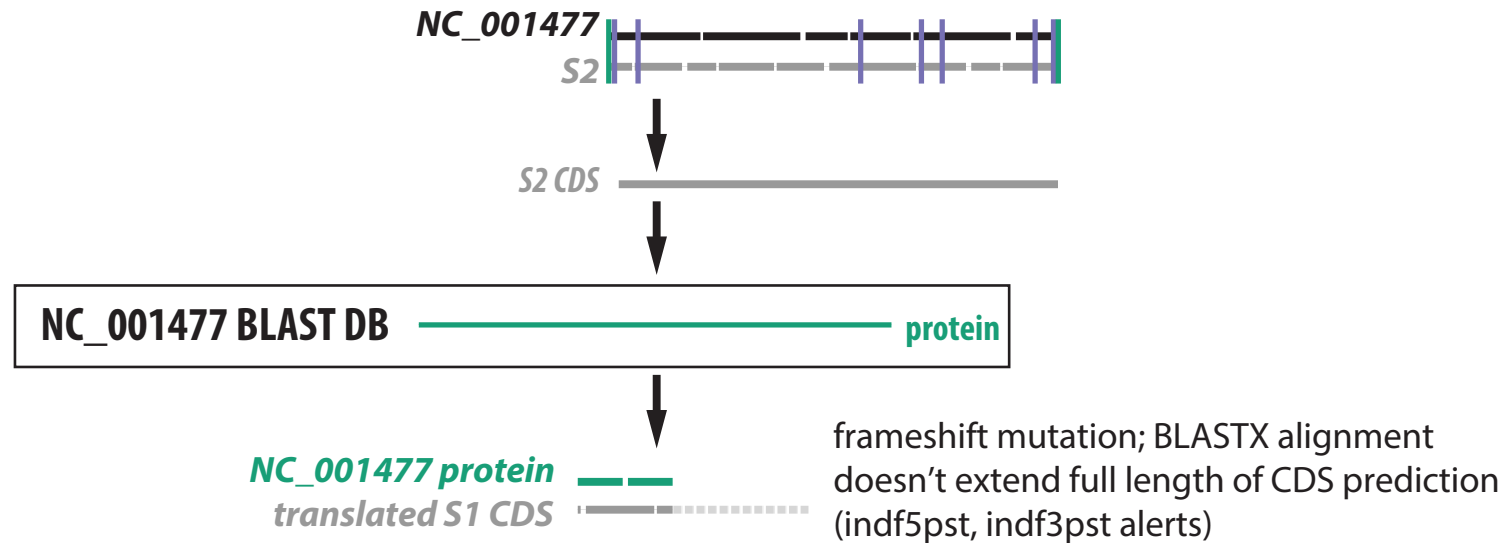
Stage 4: Protein validation (Alejandro Schäffer)

Compare each predicted CDS to model (RefSeq) proteins with BLASTX



Stage 4: Protein validation (Alejandro Schäffer)

Compare each predicted CDS to model (RefSeq) proteins with BLASTX



code	S/F	error message	description
Fatal alerts detected in the protein validation stage			
cdsstopp	F	CDS_HAS_STOP_CODON	stop codon in protein-based alignment
indfantp	F	INDEFINITE_ANNOTATION	protein-based search identifies CDS not identified in nucleotide-based search
indf5plg	F	INDEFINITE_ANNOTATION_START	protein-based alignment extends past nucleotide-based alignment at 5' end
indf5pst	F	INDEFINITE_ANNOTATION_START	protein-based alignment does not extend close enough to nucleotide-based alignment 5' endpoint
indf3plg	F	INDEFINITE_ANNOTATION_END	protein-based alignment extends past nucleotide-based alignment at 3' end
indf3pst	F	INDEFINITE_ANNOTATION_END	protein-based alignment does not extend close enough to nucleotide-based alignment 3' endpoint
indfstrp	F	INDEFINITE_STRAND	strand mismatch between protein-based and nucleotide-based predictions
insertnp	F	INSERTION_OF_NT	too large of an insertion in protein-based alignment
deletinp	F	DELETION_OF_NT	too large of a deletion in protein-based alignment

VADR used for Norovirus and Dengue virus sequences since 2018

	Norovirus	Dengue virus
length	7.6Kb	10.7Kb
# seqs	44,936	113,211
% seqs full length	5.1%	8.4%
% Ns	0.5%	0.2%
% seqs with stretch of ≥ 50 Ns	1.0%	0.4%
average % identity	81.6%	94.4%

VADR v1.0 performance

seconds per sequence	42.4	92.6
required RAM	8Gb	8Gb
total running time, CPU days	1.1	10.2

SARS-CoV-2 sequence submissions have increased since early 2020

month	year	#new seqs	#cumulative seqs
Jan	2020	32	32
Feb	2020	58	90
Mar	2020	332	422
Apr	2020	1541	1963
May	2020	2974	4937
Jun	2020	3394	8331
Jul	2020	3604	11,935
Aug	2020	3818	15,753
Sep	2020	6731	22,484
Oct	2020	11,939	34,423
Nov	2020	4274	38,697
Dec	2020	4530	43,227
Jan	2021	8775	52,002
Feb	2021	26,078	78,080
Mar	2021	42,607	120,687
Apr	2021	97,095	217,782
May	2021	104,729	322,511
Jun	2021	46,187	368,698
Jul	2021	43,336	412,034
Aug	2021	141,958	553,992
Sep	2021	267,562	821,554
Oct	2021	239,296	1,060,850
Nov	2021	267,270	1,328,120
Dec	2021	288,771	1,616,891
Jan	2022	258,522	1,875,413
Feb	2022	230,185	2,105,598

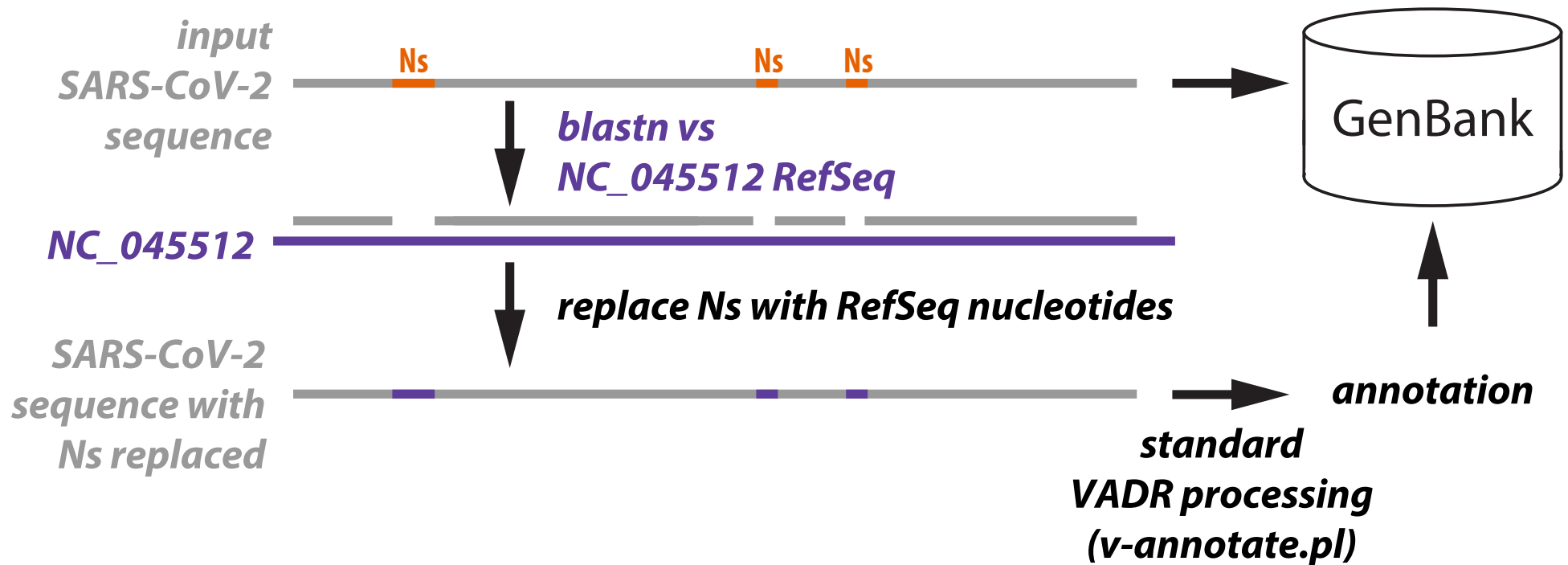
SARS-CoV-2 sequences differ from Norovirus and Dengue virus in several ways that impact VADR processing

	Norovirus	Dengue virus	SARS-CoV-2
length	7.6Kb	10.7Kb	29.9Kb
# seqs	44,936	113,211	1,616,891
% seqs full length	5.1%	8.4%	99.7%
% Ns	0.5%	0.2%	1.4%
% seqs with stretch of ≥ 50 Ns	1.0%	0.4%	38.7%
average % identity	81.6%	94.4%	99.4%

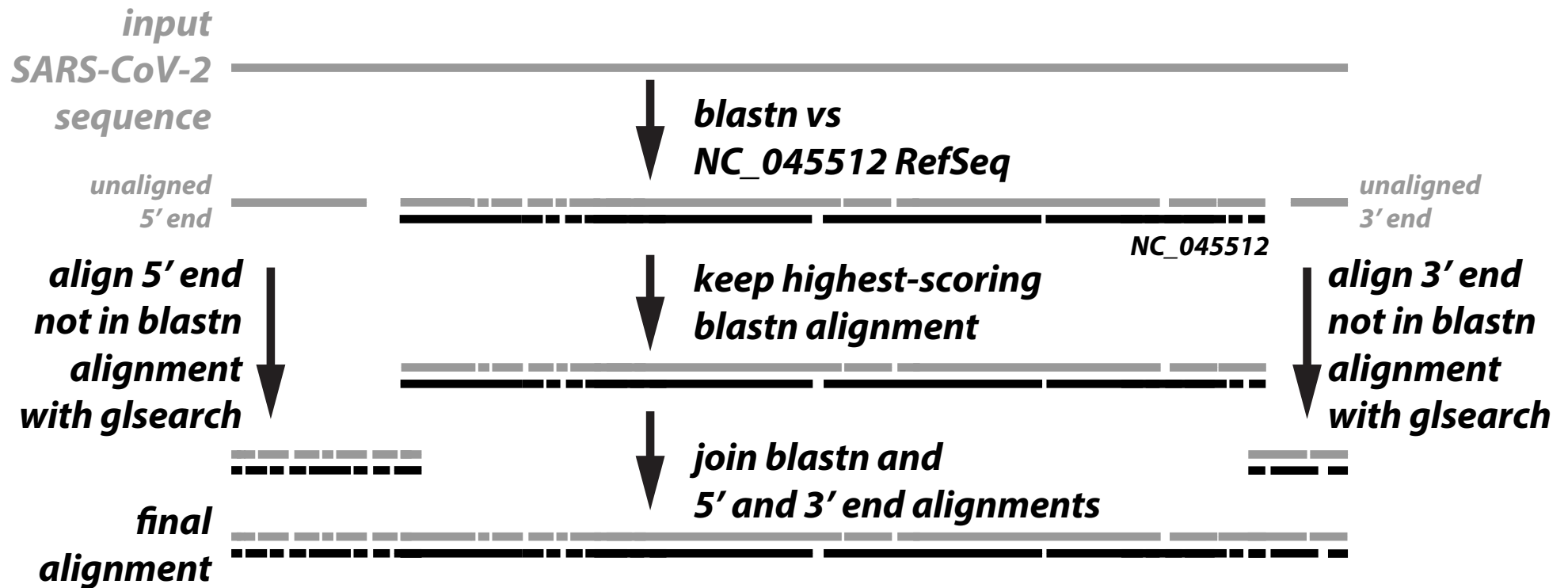
VADR v1.0 performance

seconds per sequence	42.4	92.6	331.8
required RAM	8Gb	8Gb	64Gb
total running time, CPU days	1.1	10.2	6187.6

Replacing Ns with expected nucleotides allows
many 'good' sequences to pass

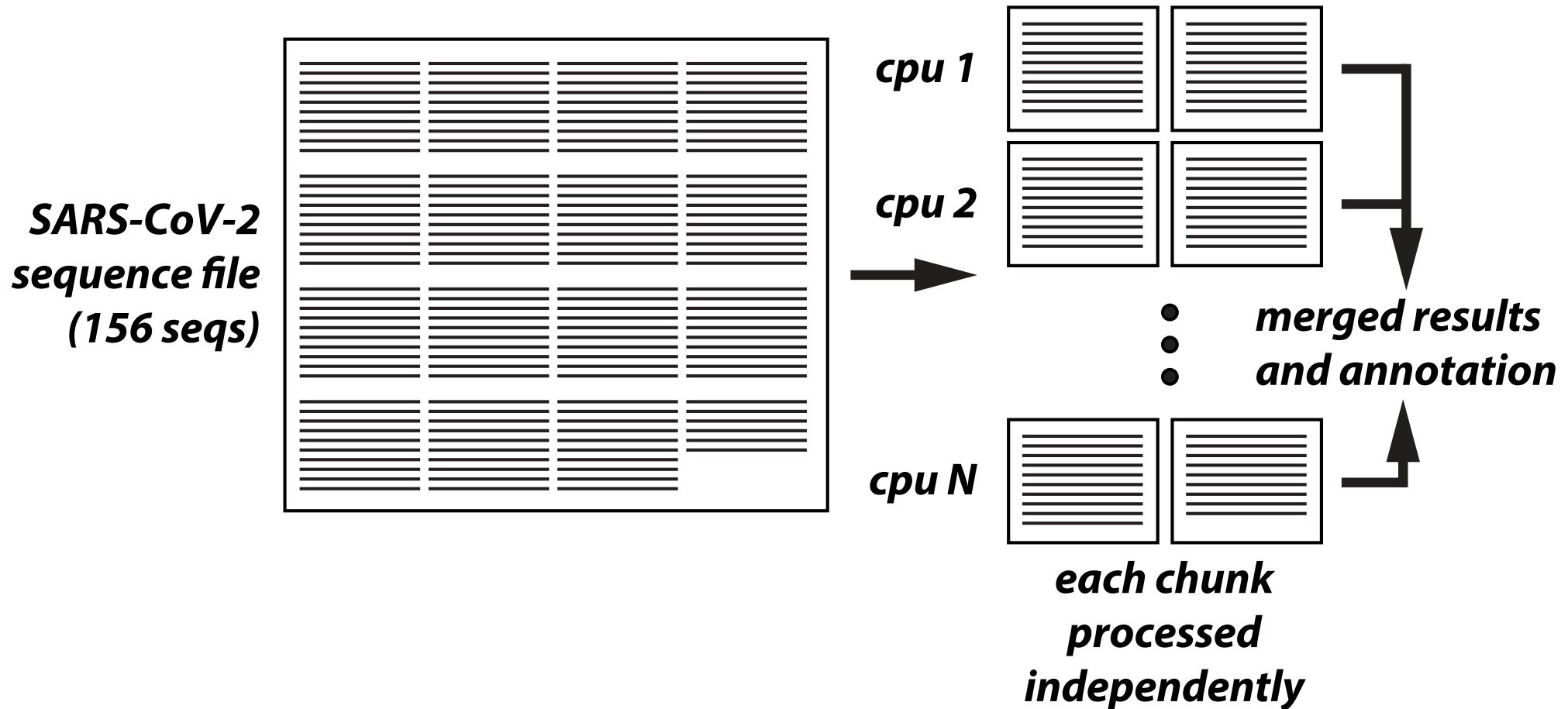


Seeded alignment using blastn makes alignment stage faster



Using glsearch instead of cmaln reduces memory requirement

- lower memory requirement (2Gb max) allows for multi-threading



VADR is now 1000-fold faster in practice for SARS-CoV-2 processing

VADR version	seeded align- ment?	N replace- ment?	glsearch?	# cpus	required RAM	secs per seq	hours per 100K seqs	speedup vs v1.0
v1.0	—	—	—	1	64 Gb	329.91	9164.3	-
v1.4.1	+	+	+	8	16 Gb	0.33	9.3	986.8

VADR is now fast enough to handle hundreds of thousands of sequences per month

month	year	#new seqs	#cumulative seqs
Jan	2020	32	32
Feb	2020	58	90
Mar	2020	332	422
Apr	2020	1541	1963
May	2020	2974	4937
Jun	2020	3394	8331
Jul	2020	3604	11,935
Aug	2020	3818	15,753
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Jan	2022	258,522	1,875,413
Feb	2022	230,185	2,105,598

**Besides getting faster, VADR has changed in other ways
(work with Linda Yankie and Vince Calhoun and GenBank team)**

- 13 releases between March 2020 and January 2022
- 3 additional models (all eventually dropped):
 - B.1.1.7 (alpha)
 - B.1.525
 - 28254-deletion
- allow some alerts for non-essential ORFs without failing sequence
(they become a `misc_feature` instead)

Faster SARS-CoV-2 sequence validation and annotation for GenBank using VADR

Eric P. Nawrocki *

National Center for Biotechnology Information, U.S. National Library of Medicine, National Institutes of Health,
Bethesda, MD, 20894, USA

Received September 08, 2022; Revised November 28, 2022; Editorial Decision December 14, 2022; Accepted January 03, 2023

Additional VADR models and development

	length	num models	new feature(s)	author
RSV	15Kb	2	alignment-based models	Eric Nawrocki
COX-1	1.5Kb	86	protein-coding gene	Eric Nawrocki
Mpox	197Kb	1	minimap alignment	Eric Nawrocki
Influenza	1-2Kb	70	segmented virus	Eric Nawrocki
Hepatitis B	1-2Kb	1	circular genome	Eric Nawrocki
Zika	11Kb	3	-	EB Dickinson

Additional VADR models and development

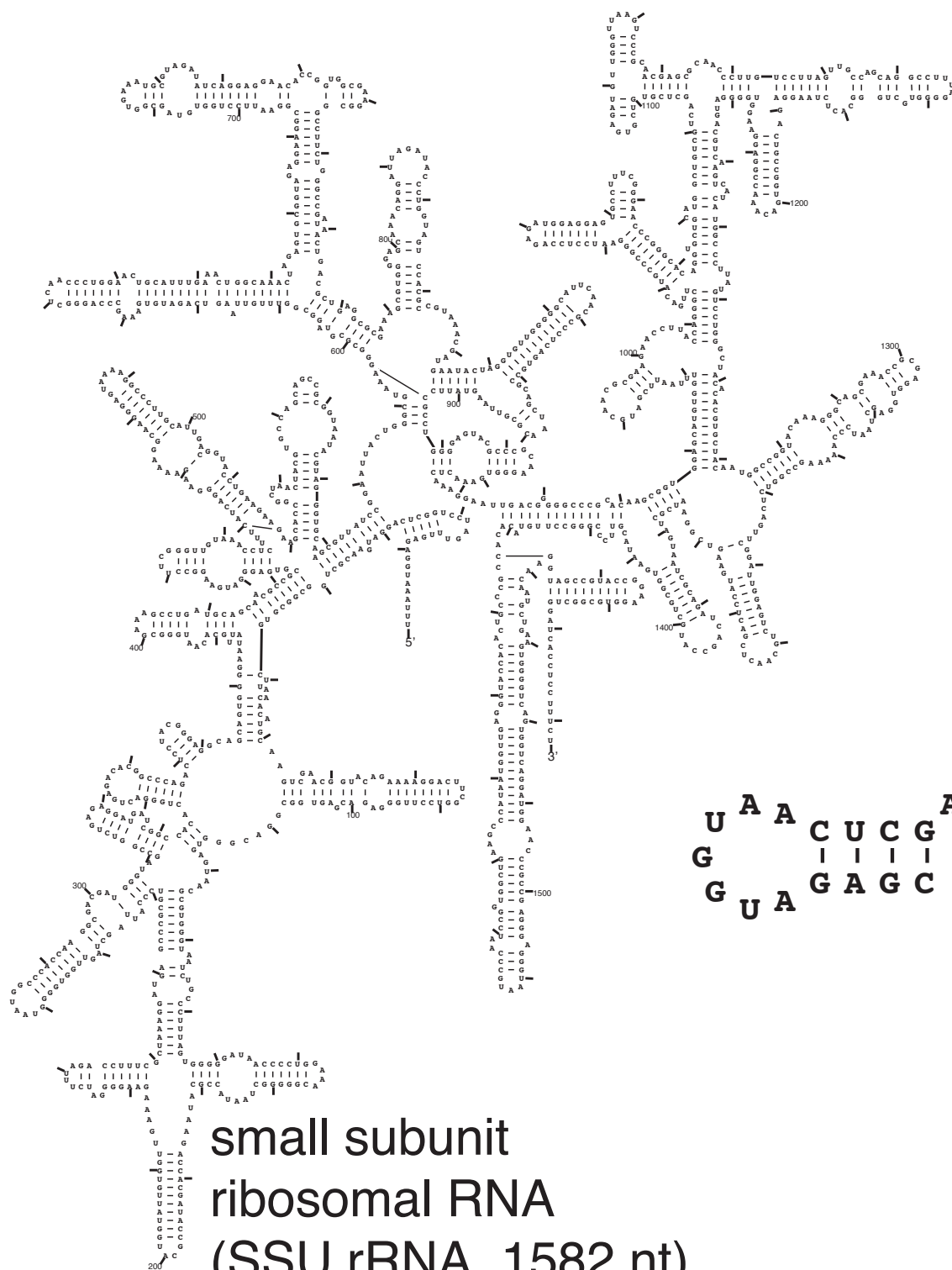
- Alex Greninger's lab at Univ of Washington:
 - sequences a wide variety of human pathogenic viruses
 - previously developed the VAPiD software tool for validating and annotating viral sequences
 - now a collaborator that builds VADR models

Additional VADR models and development

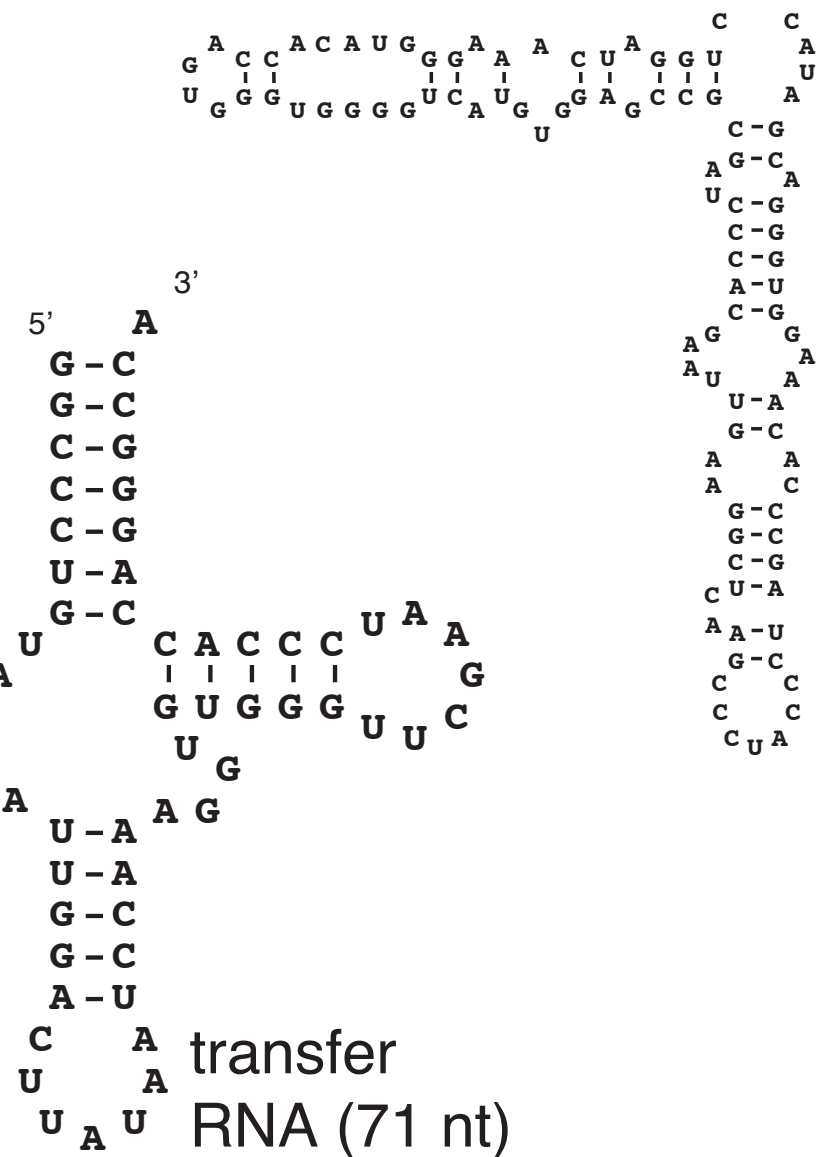
	length	num models	new feature(s)	author
Hepatitis A virus (HAV)	7Kb	5	-	Jeffrey Furlong (Greninger Lab)
Measles virus (MeV)	16Kb	15	-	Jeffrey Furlong (Greninger Lab)
Mumps virus (MuV)	15Kb	12	-	Jeffrey Furlong (Greninger Lab)
Rubella virus (RuV)	10Kb	2	-	Jeffrey Furlong (Greninger Lab)
Seasonal human coronavirus (HCoV)	30Kb	4	-	Jeffrey Furlong (Greninger Lab)
Herpes Simplex Virus (HSV)	150Kb	2	-	Jaydee Sereewit (Greninger Lab)
Human meta-pneumovirus (HMPV)	13Kb	6	-	Jeffrey Furlong (Greninger Lab)
Human para-influenza virus (HPIV)	15Kb	5	-	Jeffrey Furlong (Greninger Lab)

Future directions for VADR

- NCBI Virus FY2026 goals:
 - VADR web server
 - Replacement of FLAN with VADR
- Models for more viruses (our group and Greninger lab)
- Handling of circular genomes (Hep B)
- Alignment-based models
 - nearest-neighbor based classification of serotypes
 - possible recombination detection
- RNA structure annotation in Flaviviruses (already exists for Dengue)
- Acceleration due to speedups in Infernal



5S ribosomal RNA (119 nt)



Functional RNAs play many vital roles in the cell

	key RNAs involved	archaea	bacteria	eukarya	viruses
translation	ribosomal RNAs	x	x	x	
	transfer RNAs	x	x	x	
	RNase P RNA	x	x	x	
	snoRNAs	x		x	
	SRP RNA	x	x	x	
	tmRNA		x		
	RNaseMRP			x	
gene expression	riboswitches	?	x	?	
	microRNAs			x	x
	6S RNA		x	x	
splicing	U1, U2, U4, U5, U6			x	
other	tracrRNA	x	x		
	telomerase RNA			x	
	group I introns	x	x	x	x
	sfRNAs				x
	many more...				

Functional RNAs play many vital roles in the cell

	key RNAs involved	archaea	bacteria	eukarya	viruses
translation	ribosomal RNAs	x	x	x	
	transfer RNAs	x	x	x	
	RNase P RNA	x	x	x	
	snoRNAs	x		x	
	SRP RNA	x	x	x	
	tmRNA		x		
	RNaseMRP			x	
gene expression	riboswitches	?	x	?	
	microRNAs			x	x
	6S RNA		x	x	
splicing	U1, U2, U4, U5, U6			x	
other	tracrRNA	x	x		
	telomerase RNA			x	
	group I introns	x	x	x	x
	sfRNAs				x
	many more...				



database of more than 4100 non-coding RNA families
each represented by a secondary structure, alignment, and covariance model.

Discovery of Archaeal Group I Introns Using Infernal^a

Eric P. Nawrocki(1), Thomas A. Jones(2), and Sean R. Eddy(2)

1: National Center for Biotechnology Information, U.S. National Library of Medicine, Bethesda, MD 20894, USA.

2: Howard Hughes Medical Institute, FAS Center for Systems Biology, John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, Massachusetts 02138, USA.

nawrocke@ncbi.nlm.nih.gov

^aPoster 39 is also related to Infernal



RNA homology searches based on sequence and structure

Finding homologs of structural RNAs is challenging because the sequences are often short (100-200 nt), lack ORFs, and have regions of high sequence variability even while conserving their three-dimensional structure. The conserved sequence and secondary structure of RNAs offers two statistical signals that can be harnessed when searching databases for homologs using CMs.

In Figure 1 below, the amount of information, measured in *bits*, inherent in a sequence-only profile (14 bits) and a sequence-and-structure profile (17 bits) is shown for a toy example of an RNA family. We expect a match to a sequence-only profile for this family once in every $2^{14} = 16,384$ random nucleotides. Additionally modeling structure with a sequence-and-structure based profile (like a CM) reduces this probability 8-fold, to once every $2^{17} = 131,072$ random nucleotides.

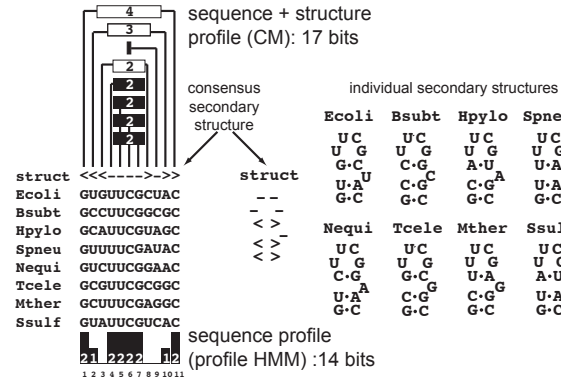


Figure 1: Information in a sequence-only versus a sequence and structure profile. Boxes with internal numbers at top and bottom of the alignment indicate the number of bits per position from the sequence (black), or per basepair from the structure (white). This figure is from [1].

The amount of additional information gained from structure varies widely for real RNA families, as shown for about 160 families in Figure 2 below. Note that for most families, modeling structure contributes at least 10 additional bits of information, which corresponds to lowering the expected chance of a false positive in a random database (i.e. the E-value of a database hit) by three orders of magnitude ($2^{10} = 1024$).

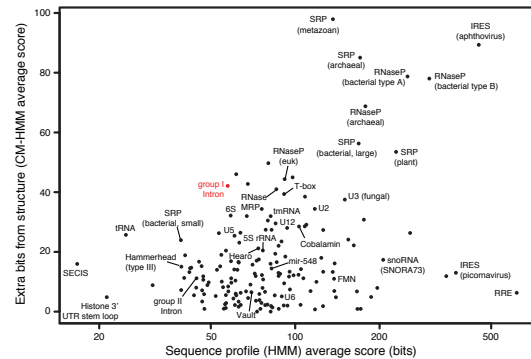


Figure 2: Additional information (in bits) gained by sequence and structure profiles (CMs) versus sequence-only profiles (HMMs) for various RNA families. Data shown for the 164 Rfam release 11.0 families with 50 or more sequences in the seed alignment plus the group I intron model (RF00028, 12 sequences). For each family, the seed alignment was used to build two profile models, a CM and a profile HMM. From each model, 10,000 sequences were generated and scored, and the average score per sampled sequence was calculated. Infernal version 1.1 was used for all steps. This figure is very similar to one from [2].

Internal benchmark shows benefit of modeling sequence and structure conservation

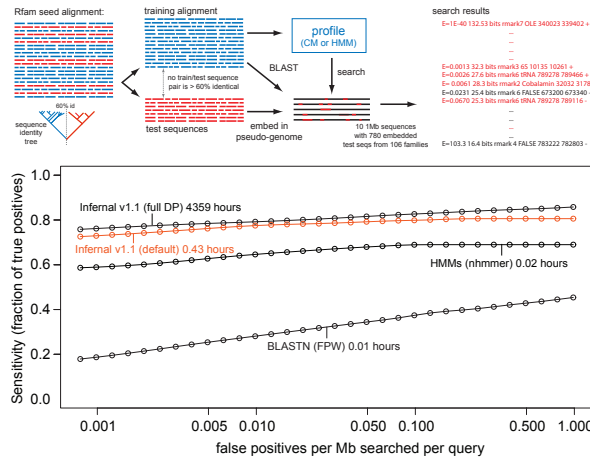


Figure 3: RMARK benchmark. Top: schematic for benchmark construction. Bottom: Results of benchmark. Plots are shown for the new Infernal 1.1 with and without filters, for profile HMM searches with nhmmer [3] (from the HHMER package included in Infernal 1.1, default parameters) and for family-pairwise-searches with BLASTN (ncbi-blast-2.2.28+ default parameters). The Infernal times do not include time required for model calibration. This figure is from [4].

Rfam: the RNA families database [5]

Rfam contains 2791 families, each represented by a

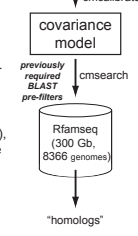
- SEED alignment
- covariance model (CM)
- list of annotated hits in Rfamseq database (more than 2 million hits)

If you submit a family to Rfam, these tools become available to the community, allowing it to be annotated in genomes and other datasets.

Rfam also includes community annotation of families (wikipedia), RNA motifs, secondary structure diagrams, alignment statistics, taxonomic information, and sequence search capability.

rfam.xfam.org

Rfam SEED alignment including consensus structure annotation



Types of RNAs in Rfam

count	type
239	Cis-reg;
28	Cis-reg; frameshift-element;
32	Cis-reg; IRES;
30	Cis-reg; leader;
31	Cis-reg; riboswitch;
10	Cis-reg; thermoregulator;
76	Gene;
29	Gene; antisense;
11	Gene; antisyn;
64	Gene; CRISPR;
217	Gene; lncRNA;
530	Gene; miRNA;
23	Gene; ribozyme;
15	Gene; rRNA;
3	Gene; snRNA;
463	Gene; snRNA; snRNA; CD-box;
266	Gene; snRNA; snRNA; HACA-box;
24	Gene; snRNA; snRNA; scaRNA;
15	Gene; snRNA; splicing;
471	Gene; sRNA;
2	Gene; tRNA;
9	Intron;

Infernal, without BLAST filters, finds new homologs

Table 1. Comparison of the old Rfam 11.0 BLAST and Infernal 1.0 search strategy versus the new Rfam 12.0 Infernal 1.1 search strategy for 15 of 200 randomly chosen families

Accession	Family ID	Length (nt)	# of seed seqs	Time new (h)	Time old (h)	Time old/new	New total hits	Old total hits	New unique hits	Old unique hits
Top five families										
RF00028	Intron apt.	251	12	125.0	357.2	2.8	71,433	60,264	11,175	1
RF00026	L16	188	18	31.2	181.1	5.8	66,517	62,174	4,343	14
RF00003	U1	166	100	11.6	64.0	5.5	15,770	14,867	904	1
RF00162	SAM	108	433	8.3	590.0	70.8	4,905	4,797	108	0
RF00050	FMN	140	144	17.1	169.9	23.9	4,081	4,066	16	1

Group I introns

Group I introns are self-splicing ribozymes found in lower eukaryotes, higher plants, bacteria and viruses. Group Is exhibit a sporadic phylogenetic distribution that is consistent with horizontal gene transfer and they often encode homing endonuclease genes which are responsible for their mobility. They are found within highly conserved genes, such as ribosomal RNAs. They are quickly evolving at the sequence level, while maintaining a core secondary structure that is crucial to their catalytic activity. Previous to this work [6] Group I introns had not been discovered in Archaea.



The Group I Intron Sequence and Structure Database [7] includes alignments and consensus structures of 15 subtypes of group I introns.

name	# seq	avglen	%id	# bps	# sig	HMM	CM	name	# seq	avglen	%id	# bps	# sig	HMM	CM
RF00028	12	364.8	34	61	8	62.6	41.7	IC1	837	436.0	39	103	84	130.3	40.3
IA1	76	583.6	45	82	51	166.5	38.0	IC2	32	320.2	66	86	27	298.1	51.7
IA2	15	276.8	38	67	24	58.0	47.5	IC3	328	255.8	67	58	16	244.1	13.2
IA3	56	282.3	46	81	50	122.4	39.1	ID	17	242.5	53	66	16	122.8	45.5
IB1	42	298.0	72	87	9	320.4	51.8	IE1	38	362.2	60	95	19	268.0	44.8
IB2	18	242.2	39	65	27	57.1	39.8	IE2	56	399.9	55	112	38	250.9	47.4
IB3	7	277.7	52	72	10	98.5	60.3	IE3	110	405.9	57	119	51	293.4	45.7
IB4	89	282.3	44	72	43	108.3	33.2								

Table 1: Attributes of the alignments of the 14 GISSD group I subtypes and models, plus the Rfam RF00028 model.

Searches for archaeal group I introns using Infernal

- downloaded all archaeal sequences in GenBank (6.7Gb as of Sept 2017)
- searched archaeal sequences with all GISSD models + RF00028 with default cm-search parameters and with -anytrunc
- 95 non-overlapping hits with $E < 0.01$ corresponding to 39 group I intron candidates (12 IA3 and 27 IB4)
- 30/39 introns have at least one hit with $E < 10^{-10}$
- 36 within LSU rRNA, 3 within SSU rRNA

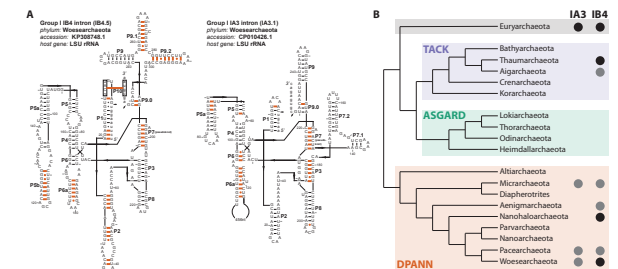


Figure 4: A) Predicted secondary structure of example archaeal group I IB4 and IB3 introns. B) Archaeal phyla in which candidate group I introns were found.

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